MILITARY SPECIFICATION

CONTAINER, SHIPPING AND STORAGE, ATOMIC PROJECTILE: M500

1. SCOPE.

1.1 Scope. This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of container used with M440, M422, M423, 8-inch atomic projectiles.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-26860 -------- Humidity Color Change

STANDARDS

MILITARY

MIL-STD-105 ------- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 ------ Ammunition Data Cards
MIL-STD-1168 ------ Lot Numbering of Ammunition
MIL-STD-1169(MU) -- Packaging, Packing and Marking for Shipments of Inert Ammunition Components
MS35338-45 -------- Washer, Lock
MS51967-5 --------- Nut, Plain, Hexagon
MS90725-34 and 36 - Bolt, Machine

FSC 8140

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DRAWINGS

ORDNANCE CORPS

8787242 ----- Cover
8791666 ----- Indicator, Humidity, Plug
8795690 ----- Shipping and Storage Container Atomic Projectile: M500
8795691 ----- Body
8795692 ----- Bolt, Machine
8795699 ----- Flange and Nut Assembly
8795702 ----- Flange
8806590 ----- Wrench, Humidity Indicator Housing H4076
8835127 ----- Cover, Atomic Projectile Shipping and Storage Container
8859012 ----- Finishing Requirements for Ferrous Metal Containers
9212613 ----- Body Assembly
ET-8795690 -- Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The container shall comply with all requirements specified on the drawing (dwg) 8795690, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Workmanship.

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8795690) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

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Table I

Inspection or Test	Contractors Inspection Permissible Before Submission	Non Dest	Parts	Sub. Assy	Assy
1. Classification of Characteristics (see 4.3.2.1)	X	X	X	X	X
2. Humidity Color Change (see 4.3.3.1)	X	X		X	
3. Pressure (see 4.3.3.2)	X	X			X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristics inspections are performed.

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Table II

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	114	106	169-170	141	253-255	176
73	72	115	107	171-172	142	256-259	177
74	73	116-117	108	173-174	143	260-262	178
75	74	118	109	175-176	144	263-265	179
76	75	119	110	177-178	145	266-268	180
77	76	120-121	111	179-180	146	269-271	181
78	77	122	112	181-182	147	272-275	182
79	78	123-124	113	183-184	148	276-278	183
80	79	125	114	185-187	149	279-282	184
81-82	80	126-127	115	188-189	150	283-285	185
83	81	128	116	190-191	151	286-289	186
84	82	129-130	117	192-193	152	290-293	187
85	83	131	118	194-195	153	294-296	188
86	84	132-133	119	196-197	154	297-300	189
87	85	134	120	198-200	155	301-304	190
88-89	86	135-136	121	201-202	156	305-308	191
90	87	137	122	203-204	157	309-312	192
91	88	138-139	123	205-207	158	313-316	193
92	89	140-141	124	208-209	159	317-320	194
93	90	142	125	210-212	160	321-324	195
94-95	91	143-144	126	213-214	161	325-329	196
96	92	145	127	215-217	162	330-333	197
97	93	146-147	128	218-219	163	334-338	198
98	94	148-149	129	220-222	164	339-342	199
99-100	95	150	130	223-224	165	343-347	200
101	96	151-152	131	225-227	166	348-352	201
102	97	153-154	132	228-230	167	353-356	202
103	98	155-156	133	231-232	168	357-361	203
104-105	99	157	134	233-235	169	362-366	204
106	100	158-159	135	236-238	170	367-372	205
107	101	160-161	136	239-241	171	373-377	206
108-109	102	162-163	137	242-243	172	378-382	207
110	103	164-165	138	244-246	173	383-388	208
111	104	166-167	139	247-249	174	389-393	209
112-113	105	168	140	250-252	175	394-399	210

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Table II (Continued)

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
400-405	211	531-540	229	752-768	247	1205-1243	265
406-411	212	541-549	230	769-785	248	1244-1285	266
412-417	213	550-559	231	786-804	249	1286-1330	267
418-424	214	560-569	232	895-823	250	1331-1378	268
425-430	215	570-579	233	824-842	251	1379-1429	269
431-437	216	580-590	234	843-863	252	1430-1483	270
438-443	217	591-601	235	864-884	253	1484-1542	271
444-450	218	602-613	236	885-907	254	1543-1605	272
451-457	219	614-625	237	908-930	255	1606-1673	273
458-465	220	626-637	238	931-955	256	1674-1747	274
466-472	221	638-649	239	956-981	257	1748-1827	275
473-480	222	650-662	240	982-1008	258	1828-1914	276
481-488	223	663-676	241	1009-1036	259	1915-2010	277
489-496	224	677-690	242	1037-1066	260	2011-2115	278
497-504	225	691-704	243	1067-1098	261	2116-2232	279
505-513	226	705-719	244	1099-1131	262	2233-2361	280
514-521	227	720-735	245	1132-1166	263	over 2361	298
522-530	228	736-751	246	1167-1204	264		

Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Flange (see dwg 8795702) covering a detail of dwg 8795699.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Location of holes	Gage
102.	Diameter of holes	Gage
103.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.2 Flange and nut assembly (see dwg 8795699) covering a detail of dwg 9212613.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Pitch diameter of thread (see MS51967-5)	Gage
102.	Minor diameter of thread (see MS51967-5)	Gage
103.	Welding	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.3 Body (see dwg 8795691) covering a detail of dwg 9212613.

Categories	Characteristics	Method of Inspection
Critical: None defined		

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Major
101. Total length ..... Gage
102. Material (see 3.1) ..... Visual

Minor
201. Workmanship (see 3.4) ..... Visual/Manual

4.3.2.1.4 Body assembly (see dwg 9212613) covering a detail of dwg 8795690.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Location of rings	*Measure
102.	Location of handles	*Measure
103.	Location of flange	Gage
104.	Welding in accordance with dwg 9212613	Visual/Manual
105.	Inside diameter (see 8795691)	Gage
106.	Objective evidence (see 6.2) for protective finish and cleaning in accordance with dwg 8859012	Visual
107.	No paint on thread	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

*Measure with suitable scale.

4.3.2.1.5 Housing, card humidity indicator (see dwg 8791682) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Large inside diameter adjacent to thread, min.	Gage
102.	Small diameter of groove in flange, max.	Gage
103.	Width of groove in flange, min.	Gage

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104.	Depth of groove in flange, max.	Gage
105.	Width of small inside groove, min.	Gage
106.	Small diameter of small inside groove, max.	Gage
107.	Depth of small inside groove, max.	Gage
108.	Material (see 3.1)	Visual
109.	Objective evidence (see 6.2) for finish in accordance with dwg 8791682	Visual
Minor		
201.	Surface finish	Visual
202.	Stamping	Visual
203.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.6 Wrench, humidity indicator housing H4076 (see dwg 8806590) covering a detail of dwg 8791666.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Thickness	Gage
102.	Depth of notches, min.	Gage
103.	Length of tab	Gage
104.	Material (see 3.1)	Visual
105.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8806590	Visual
Minor		
201.	Stamping	Visual
202.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.7 Indicator humidity plug (see dwg 8791666) covering a detail of dwg 8835127.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	No part missing or improperly assembled	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

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4.3.2.1.8 Cover (see dwg 8787242) covering a detail of dwg 8835127.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Pitch diameter of threaded holes	Gage
102.	Minor diameter of threaded holes	Gage
103.	Diameter of large hole	Gage
104.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.9 Cover, atomic projectile, shipping and storage container (see dwg 8835127) covering a detail of dwg 8795690.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Torque humidity indicator, min.	Gage
102.	Chain brazed to cover	Visual
103.	Objective evidence (see 6.2) for protective finish and cleaning in accordance with dwg 8859012	Visual
104.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.10 Bolt, machine (see dwg 8795692) covering a detail of dwg 8795690.

Categories	Characteristics	Method of Inspection
Critical: None defined		

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Major
101. Objective evidence (see 6.2) for screw in accordance with MS90725-36 ..... Visual
102. Diameter of hole, min. ..... Gage

Minor
201. Workmanship (see 3.4) ..... Visual/Manual

4.3.2.1.11 Shipping and storage container, atomic projectile: M500 (see dwg 8795690).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Objective evidence (see 6.2) for washer, lock in accordance with MS35338-45	Visual
102.	Objective evidence (see 6.2) for screw, cap in accordance with MS90725-34	Visual
103.	No part missing or improperly assembled	Visual
104.	Material (see 3.1)	Visual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.12 Preparation for delivery (see section 5).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Preservation, packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual/Manual
Minor		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

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Test	Test Classification
a. Humidity color change	Major
b. Pressure	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Humidity color change (see dwg 8791666). Each indicator, humidity, plug shall be tested in accordance with 4.4.1. If any item fails to comply with drawing requirements, the item shall be rejected and removed from the lot.

4.3.3.2 Pressure (see dwg 8795690). Each container shall be tested in accordance with 4.4.2. If any item fails to comply with drawing requirements, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8795690. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Humidity color change (see 4.3.3.1). Each indicator, humidity, plug shall be tested in accordance with MIL-I-26860.

4.4.2 Pressure (see 4.3.3.2). Each container shall be pressurized in accordance with dwg 8795690. Turn off pressure supply and observe gage for any indication of drop in pressure for the length of time specified.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The container shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The container shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The container shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 8140-A083

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-50565(MU) Container, Shipping and Storage, Atomic Projectile: M500

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIR INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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